

Note about Green functions and the Newns-Anderson model

Karsten W. Jacobsen, Mads Brandbyge

Abstract

In this note we introduce the concept of Green functions in quantum mechanics. We focus on one-body Green functions and their use as exemplified by the Newns-Anderson model. We discuss how the concept can be extended to the many-body case.

In the following we let $\hat{H}$ denote a one-body Hamiltonian operator and we assume that $|\alpha\rangle$ is a complete set of orthonormal eigenstates of $\hat{H}$ with corresponding eigenvalues ε_α :

$$\hat{H}|\alpha\rangle = \varepsilon_\alpha|\alpha\rangle. \quad (1)$$

Now assume that we can divide up $\hat{H}$ in two parts, $\hat{H} = \hat{H}_0 + \hat{V}$, and let $|i\rangle$ denote an orthonormal basis of eigenstates for the “unperturbed” Hamiltonian operator $\hat{H}_0$:

$$\hat{H}_0|i\rangle = \varepsilon_i|i\rangle. \quad (2)$$

(We shall use latin letters $i, j \dots$ to denote eigenstates of $\hat{H}_0$ and greek letters $\alpha, \beta, \dots$ to denote eigenstates of $\hat{H}$.)

We will focus on a particular eigenstate state for the uncoupled system, i , and will assume that the only non-zero coupling is for $\langle i|\hat{V}|j\rangle$ where i is our chosen index and $j \neq i$. Thus the diagonal elements of $\hat{V}$ vanishes, $\langle i|\hat{V}|i\rangle = 0$, so that $\varepsilon_i = \langle i|\hat{H}|i\rangle$ (other wise it could just be included by re-defining $\varepsilon_i \rightarrow \varepsilon_i + \hat{V}_{ii}$), and $\langle j|\hat{V}|j'\rangle = 0$ for $j, j' \neq i$.

We now aim at calculating the *local density of states* $\rho_i(\varepsilon)$ at the state $|i\rangle$ for the system with the Hamiltonian $\hat{H}$. The definition of the *local* or *projected* density of states, $\rho_i(\varepsilon)$, is

$$\rho_i(\varepsilon) = \sum_{\alpha} |\langle i|\alpha\rangle|^2 \delta(\varepsilon - \varepsilon_\alpha) = \langle i|\delta(\varepsilon - \hat{H})|i\rangle. \quad (3)$$

The last expression with a δ function of an operator can be defined through the spectral decomposition

$$\delta(\varepsilon - \hat{H}) = \sum_{\alpha} |\alpha\rangle \delta(\varepsilon - \varepsilon_\alpha) \langle \alpha|. \quad (4)$$

The Green operator and the Green function

In order to calculate the density of states we introduce the Green operator

$$\hat{G}(t) = -i \exp(-i\hat{H}t)\theta(t), \quad (5)$$

where θ denotes the step function (the Heaviside function), which is one for positive times and zero for negative times. The physical meaning of the Green operator is that it describes the time evolution for positive times. The Fourier transform of $\hat{G}$ is given by

$$\hat{G}(\varepsilon) = \lim_{\eta \rightarrow 0^+} \frac{1}{\varepsilon + i\eta - \hat{H}} = \frac{P}{\varepsilon - \hat{H}} - i\pi\delta(\varepsilon - \hat{H}). \quad (6)$$

In this equation P denotes the Cauchy principal value.

We now define the *Green function* $G_i(z)$ of the state i as $G_i(z) = \langle i|\hat{G}(z)|i\rangle$ and see that the local density of states (3) can be obtained as

$$\rho_i(\varepsilon) = -\frac{1}{\pi} \text{Im} G_i(\varepsilon) = -\frac{1}{\pi} \text{Im} \langle i|\hat{G}(\varepsilon)|i\rangle. \quad (7)$$

So we see, that if we know the Green operator $\hat{G}$ or more specifically the Green function $G_i(\varepsilon)$ of the state i , we can get the local density of states. We therefore now aim at calculating the Green function. More generally we shall consider the operator

$$\hat{G}(z) = \frac{1}{z - \hat{H}}, \quad (8)$$

where z is a complex number in the upper part of the complex plane.

Exercise 1: Show, that the Green operator can be written

$$\hat{G}(z) = \sum_{\alpha} \frac{|\alpha\rangle\langle\alpha|}{z - \varepsilon_{\alpha}}, \quad (9)$$

and that $G_i(z)$ thus can be written

$$G_i(z) = \sum_{\alpha} \frac{|\langle\alpha|i\rangle|^2}{z - \varepsilon_{\alpha}}. \quad (10)$$

Consider, that this means that all eigenvalues of the eigenvectors ($|\alpha\rangle$), in which the state $|i\rangle$ has some weight, will show up as poles in $G_i(z)$.

We can now make a series expansion of the Green operator in $\hat{V}$:

Exercise 2: Show that

$$\hat{G}(z) = \hat{G}^0(z) + \hat{G}^0(z)\hat{V}\hat{G}^0(z) + \hat{G}^0(z)\hat{V}\hat{G}^0(z)\hat{V}\hat{G}^0(z) + \dots, \quad (11)$$

where $\hat{G}^0(z) = 1/(z - \hat{H}_0)$.

With the definitions $G_i(z) = \langle i|\hat{G}(z)|i\rangle$ and $V_{ij} = \langle i|\hat{V}|j\rangle$ show that

$$G_i(z) = G_i^0(z) + \sum_{j \neq i} G_i^0(z)V_{ij}G_j^0(z)V_{ji}G_i^0(z) + \dots. \quad (12)$$

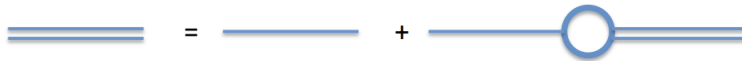
Represent this equation graphically by associating a line with label i for every unperturbed Green function $G_i^0(z)$. The full Green function is denoted by a double line. The V_{ij} 's appear between two Green functions and can be shown as a cross where the two lines with labels i and j meet. This is a simple example of what is known as Feynman diagrams.



We now introduce the self energy $\Sigma_i(z)$ as the sum over all diagrams beginning with a V_{ij} and ending with a V_{ki} which never uses G_i^0 . So we only use all the other unperturbed Green functions G_j^0 with $j \neq i$. The self energy can for example be shown graphically as a circle.

Exercise 3: Show, for example graphically, the Dyson equation

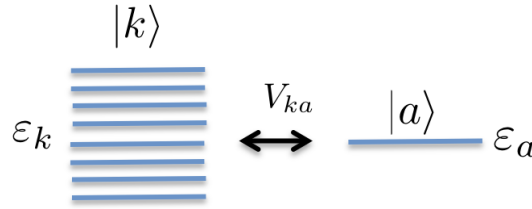
$$G_i(z) = G_i^0(z) + G_i^0(z)\Sigma_i(z)G_i(z) \quad (13)$$



Newns-Anderson model

The Newns-Anderson model describes a situation where a localized state couples to a set of possibly extended states. This could for example be an atom or molecule on a metal surface. The localized state is then a state on the atom or molecule (named the “adsorbate” state) which is coupled to all the extended states in the metal (the “metal” states). The adsorbate state is denoted $|a\rangle$ and the metal states by $|k\rangle$ where k is an index running over the metal states. The unperturbed Hamiltonian $\hat{H}_0$ describes the situations where there is no coupling between the adsorbate and the metal, so $|a\rangle$ and $|k\rangle$ are eigenstates of $\hat{H}_0$:

$$\hat{H}_0|a\rangle = \varepsilon_a|a\rangle \quad \hat{H}_0|k\rangle = \varepsilon_k|k\rangle, \quad (14)$$



Exercise 4: What does the Hamiltonian matrix look like in the basis of $|a\rangle$ and all the $|k\rangle$ -states?

The full Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}$ includes a coupling between the adsorbate state and the metal states (but not between two different metal states):

$$\langle a|\hat{V}|k\rangle = V_{ak} \quad \langle k|\hat{V}|k'\rangle = 0. \quad (15)$$

It turns out that this Hamiltonian matrix can be exactly diagonalized. However, we shall approach the problem by finding the local density of states on the adsorbate state $|a\rangle$. To do this we shall calculate the Green function $G_a(z)$.

Exercise 5: Draw the Feynman diagrams for the Newns-Anderson model and show that the self energy $\Sigma_a(z)$ of the adsorbate state is given by

$$\Sigma_a(z) = \sum_k \frac{|V_{ak}|^2}{z - \varepsilon_k}. \quad (16)$$

Show, that the Green function can then be written

$$G_a(z) = \frac{1}{z - \varepsilon_a - \Sigma_a(z)}. \quad (17)$$

To find the density of states we are interested in the limit of z going towards the real axis from above in the complex plane. Let us write $z = \varepsilon + i\eta$, where ε is real, and look at the limit where η is positive and goes to zero. The self energy can be split up in real and imaginary parts $\Sigma_a(\varepsilon) = \Lambda(\varepsilon) - i\Delta(\varepsilon)$, where

$$\Delta(\varepsilon) = \pi \sum_k |V_{ak}|^2 \delta(\varepsilon - \varepsilon_k), \quad (18)$$

and

$$\Lambda(\varepsilon) = P \sum_k \frac{|V_{ak}|^2}{\varepsilon - \varepsilon_k} = \frac{1}{\pi} P \int d\varepsilon' \frac{\Delta(\varepsilon')}{\varepsilon - \varepsilon'}. \quad (19)$$

(Bonus info: The last integral is called a Hilbert transform, so the real (Λ) and imaginary (Δ) parts of the self energy are related by a Hilbert transform. The reason this relation appears is because the Green function is defined to be zero for negative times. However, to see this one has to go more into complex function theory.)

The local density of states in the state $|a\rangle$ thus becomes

$$\rho_a(\varepsilon) = \frac{1}{\pi} \frac{\Delta(\varepsilon)}{(\varepsilon - \varepsilon_a - \Lambda(\varepsilon))^2 + (\Delta(\varepsilon))^2}. \quad (20)$$

So we found the solution! This is the exact result for the density of states projected onto the state $|a\rangle$!

You will investigate the Newns-Anderson model more in an upcoming problem set. Here, we shall just look at the simple case where $\Delta(\varepsilon)$ is a constant as a function of energy $\Delta(\varepsilon) = \Delta_0$. Because the integrand in the Hilbert transform is odd the integral vanishes $\Lambda = 0$. In this simple case the local density of states then becomes a Lorentzian

$$\rho_a(\varepsilon) = \frac{1}{\pi} \frac{\Delta_0}{(\varepsilon - \varepsilon_a)^2 + \Delta_0^2}. \quad (21)$$

But hey, does this make any sense? We started with $\Delta(\varepsilon)$ as a sum of δ functions and now we assume it is a constant? Let us look into this aspect a bit more.

When is it reasonable to consider the imaginary part of the self energy $\Delta(\varepsilon)$ as a continuous function of ε ?

Let us first consider the situation with a small but finite value of η . A finite η corresponds to the situation where the step function in the Green operator in equation (5) is replaced with a function, which is zero for negative values (like the step function itself) and for positive values of the time is given by $\exp(-\eta t)$. This is a good approximation if we are considering times, which are much shorter than $1/\eta$. We can also see that for finite values of η , the δ functions appearing in the imaginary part of the self energy $\Delta(\varepsilon)$ are replaced by Lorentz functions with a typical width in energy of η . If the distance, $\delta\varepsilon$ between the energy levels ε_k in the metal is much smaller than η the spikey character of $\Delta(\varepsilon)$ disappears and it becomes a good approximation to consider this a smooth function of the energy. So to conclude, we can regard $\Delta(\varepsilon)$ as a smooth function of energy if we are considering times which are much smaller than $1/\delta\varepsilon$, where $\delta\varepsilon$ is the energy spacing in the metal.

Exercise 6: Consider a gram of copper. Knowing that the bandwidths of metals are of the order 1 Hartree, how close are the energy levels ($\delta\varepsilon$)? For how long times can we use the approximation that $\Delta(\varepsilon)$ is considered a smooth function?

To describe certain phenomena, like exponential decay of a quantum mechanical state, it is important that $\Delta(\varepsilon)$ can be considered a continuous function. Let us for simplicity look at the situation with a constant imaginary part of the self energy $\Delta(\varepsilon) = \Delta_0$, where the Green function becomes

$$G_a(\varepsilon) = \frac{1}{\varepsilon - \varepsilon_a + i\Delta_0}. \quad (22)$$

If we Fourier transform this back to real time we get

$$G_a(t) = -i \exp(-i\varepsilon_a t - \Delta_0 t) \quad t > 0. \quad (23)$$

(for negative times we get zero as in the definition Eq. (5)).

Now, consider a particle in the state $|a\rangle$ at time $t = 0$. What is the probability $P_a(t)$ that the particle is still in the same state at time t ? The answer is

$$P_a(t) = \left| \langle a | \exp(-i\hat{H}t) | a \rangle \right|^2 = |G_a(t)|^2 = \exp(-2\Delta_0 t). \quad (24)$$

So we see that we have an exponential decay with a life time of $1/(2\Delta_0)$.

The exponential decay cannot occur if we consider only a finite number of $|k\rangle$ -states. The state $|a\rangle$ can be written as a linear combination of the eigenstates $|\alpha\rangle$:

$$|a\rangle = \sum_{\alpha} |\alpha\rangle \langle \alpha | a \rangle. \quad (25)$$

If the system starts in the state $|a\rangle$ at time $t = 0$, the state at time t is given by

$$\exp(-i\hat{H}t)|a\rangle = \sum_{\alpha} \exp(-i\varepsilon_{\alpha}t) |\alpha\rangle \langle \alpha | a \rangle. \quad (26)$$

If the sum on the right hand side is finite, the system will always at some later time return to the initial state (or at least get very close to it). So no exponential decay. The trick to consider $\Delta(\varepsilon)$ to be continuous means that we limit ourselves to times short compared to $1/\delta\varepsilon$ and for these times the decay looks exponential.

As pointed out by Ballentine we cannot have exponential decay at short times, but that is related to the fact that the spectrum has a lower bound. If we assume a constant $\Delta(\varepsilon)$ we work with a model without any lower bound and we do not see the non-exponential behavior at short times.

Let us finally look at a situation where $\Delta(\varepsilon)$ should clearly *not* be considered a smooth function. The result for $G_a(z)$ (Eq. (17)) is exact and does also hold in the case where we have only a single state $|k\rangle$ with energy ε_k . As discussed earlier the eigenvalues show up as poles in the Green function so we can find them by putting the denominator of $G_a(z)$ equal to zero:

$$z - \varepsilon_a - \frac{|V_{ak}|^2}{z - \varepsilon_k} = 0 \quad (27)$$

If this equation is multiplied by $z - \varepsilon_k$ we find the usual equation to determine the eigenenergies of a two level problem with a bonding and antibonding solution.

Many-body Green functions

We have seen how Green functions/operators describing single particles can be used to treat infinite systems such as the model "chain" system. Important applications is for quantum systems coupled to "reservoirs", say an atom with discrete levels interacting with the continuum of photons or a molecule interacting with a metal. Now our focus is to show how the concept of Green functions can be extended to many-body systems treated by the second quantization formalism.

We have defined the forward-in-time propagator or retarded Green function (using atomic units: $\hbar = 1$),

$$\hat{G}(t) = -ie^{-i\hat{H}t}\theta(t) \quad (28)$$

Now this quantity is general for any system with a Hamiltonian operator and might be concerned with many particles. For a time-independent system it can be written in terms of energy eigenstates labelled by k ,

$$\hat{G}(t) = -i\theta(t) \sum_k e^{-iE_k t} |\Psi_k\rangle \langle \Psi_k| \quad (29)$$

with the Fourier/energy version,

$$\hat{G}(\varepsilon) = \lim_{\eta \rightarrow 0+} \sum_k \frac{|\Psi_k\rangle \langle \Psi_k|}{\varepsilon - E_k + i\eta} \quad (30)$$

The convenience of writing it as an operator is that it is clear that we can choose any basis to represent it. Say, we want to know how it looks in the basis $\{|i\rangle\}$,

$$G_{ij}(t) = -i\theta(t) \langle i | e^{-i\hat{H}t} | j \rangle = -i\theta(t) \sum_k e^{-iE_k t} \langle i | \Psi_k \rangle \langle \Psi_k | j \rangle. \quad (31)$$

This is the amplitude for the process where the system starts in state j at time zero and is found in state i at a later time, t .

If we consider *single particles* then this basis could represent discrete "sites" in space, or in real-space basis,

$$G(x', t; x, 0) = -i\theta(t) \langle x' | e^{-i\hat{H}t} | x \rangle = -i\theta(t) \sum_k e^{-iE_k t} \Psi_k(x') \Psi_k^*(x) \quad (32)$$

and the spin variable could be included in the index. This quantity is a complex number, an amplitude, which tell us what the amplitude is for the situation where we know the particle is at position x at time zero, and the particle reach x' at time t .

The generalization of this to the many-body case where the Hamiltonian, $\hat{H}$, and the eigenstates, Ψ_k , are describing several particles requires a bit more consideration. We will leave particle interactions to a more specialized course and here just consider the generalization to the noninteracting many-body case for fermions. A noninteracting system¹ can be described in 2nd quantization form as,

$$\hat{H}_0 = \sum_k E_k c_k^\dagger c_k \quad (33)$$

where the noninteracting ground-state (GS) is $|\Psi_0\rangle$, and the excited eigenstates $|\Psi_k\rangle = c_k^\dagger |\Psi_0\rangle$. The GS is just the filled Fermi sea,

$$|\Psi_0\rangle = \prod_{E_k < E_F} c_k^\dagger |0\rangle, \quad (34)$$

¹We put a 0-index on $\hat{H}$ since in many-body theory this is taken as the unperturbed starting-point before introducing interactions.

where $|0\rangle$ is the zero-particles/"vacuum state".

In general we express the Hamiltonian in the general single-particle basis,

$$\hat{H}_0 = \sum_{ij} h_{ij} c_i^\dagger c_j \quad (35)$$

When we, in the beginning of the note, considered systems containing just one particle, we used the basis set $|j\rangle = c_j^\dagger|0\rangle$, and defined the Green function,

$$G_{ij}(t) = -i\theta(t)\langle 0|c_i e^{-i\hat{H}_0 t} c_j^\dagger|0\rangle \quad (36)$$

We introduce the Heisenberg picture before generalizing this to the many-particle case, as this is the standard when using 2nd quantization.

Heisenberg picture

In the Schrödinger "picture" the states time-evolve according to the Schrödinger equation. This is normally used in the introduction to QM. In the alternative Heisenberg "picture" the state vectors are time-independent, while all time-dependence is placed on the operators² by a unitary transformation, so the end all matrix elements and physics will (and should, of course) be the same in the two pictures (here spelled out by the subscript H/S),

$$\langle \chi_1(t)|\hat{O}_S|\chi_2(t)\rangle_S = \langle \chi_1|e^{i\hat{H}_0 t}\hat{O}_S e^{-i\hat{H}_0 t}|\chi_2\rangle = \langle \chi_1|\hat{O}_H(t)|\chi_2\rangle_H \quad (37)$$

identifying,

$$\hat{O}_H(t) = e^{i\hat{H}_0 t}\hat{O}_S e^{-i\hat{H}_0 t} \quad (38)$$

$$|\chi\rangle_H = e^{i\hat{H}_0 t}|\chi(t)\rangle_S \quad (39)$$

The Heisenberg-picture operators fulfil the Heisenberg equation-of-motion (cf. Ballentine p. 90-91),

$$\frac{d\hat{O}_H(t)}{dt} = i[\hat{H}_0, \hat{O}_H] + \left(\frac{\partial \hat{O}}{\partial t}\right)_H \quad (40)$$

where we, for completeness, include an explicit ("Schrödinger"-like) time dependence, if the operator has one as a partial derivative. Here it is seen how all explicitly time-independent operators which are commuting with the time-independent Hamiltonian, $\hat{H}_0$, will also be time-independent. We will leave out the "H" subscript in the Heisenberg picture – it becomes too messy and it is also

²Of course the operators could themselves explicitly depend on time, for example an time-dependent B-field.

apparent when we put a time-argument put after an operator. So for example we have,

$$c_i^\dagger(t) = e^{i\hat{H}_0 t} c_i^\dagger e^{-i\hat{H}_0 t}, \quad (41)$$

which one can think of as three steps: (1) propagating the time-zero-states to time t , (2) acting with the operator, (3) propagating back to time-zero.

The time dependence of the eigenstate operator $c_k^\dagger(t)$ can be found from (40),

$$c_k^\dagger(t) = e^{iE_k t} c_k^\dagger(0) \equiv e^{iE_k t} c_k^\dagger \quad (42)$$

and by daggering $c_k(t) = e^{-iE_k t} c_k$. If we use the single-particle basis functions ϕ_k , we can write $c_k^\dagger = \sum_i c_i^\dagger \langle i|k \rangle = \sum_i \phi_k^*(i) c_i^\dagger$ (Ballentine p. 483), and get

$$c_i^\dagger(t) = \sum_k e^{iE_k t} \phi_k^*(j) c_k^\dagger. \quad (43)$$

Here the index i (j) could typically be some orbital-basis set or real-space x (x').

Exercise 7: Prove (40) using (38), and use it to show (42). You may use that for general operators A, B, C , $[A, BC] = \{A, B\}C - B\{A, C\}$ (prove this also).

Non-interacting many-body Green functions

We can replace the expression in (36) for just one particle in the Heisenberg picture by,

$$G_{ij}(t - t') = -i\theta(t - t') \langle 0 | \{c_i(t), c_j^\dagger(t')\} | 0 \rangle \quad (44)$$

where we use the anti-commutator $\{\cdot, \cdot\}$, when considering Fermions³. Here we instead of time-zero (start time $t' = 0$) explicitly put two times, t and t' . It is easy to show (try yourself!)⁴ that this definition fits the single-particle definition in (36).

It may seem strange at first to put the anti-commutator there, but for a many-particle *non-interacting* fermion system we replace the vacuum state by the non-interacting many-particle ground state, the filled "Fermi sea", $|\Psi_0\rangle$, and define the retarded Green function,

$$G_{ij}(t - t') = -i\theta(t - t') \langle \Psi_0 | \{c_i(t), c_j^\dagger(t')\} | \Psi_0 \rangle. \quad (45)$$

³For bosons we would use the normal commutator $[\cdot, \cdot]$.

⁴Use that $e^{-i\hat{H}_0 t} |0\rangle = |0\rangle$ and that $c_i(t) |0\rangle = 0$.

Here it is clear that the term where we remove a particle (create a "hole") $c_j^\dagger(t')c_i(t)|\Psi_0\rangle$ does not yield zero. With $|\Psi_0\rangle$ replacing the vacuum state we can both create particles and holes.

The interpretation of this is that we create an excitation – put a particle in state j at time t' , and remove it again from state i at a later time, t . We can equally well have the excitation process where we introduce a "hole" in state i at time t , and remove it again from state j at t' . The anti-commutator makes sure that the probability for staying in the initial state, j , obtained from the retarded Green function (amplitude) goes towards 1 when $t \rightarrow t'$, since $\{c_i(t), c_j^\dagger(t')\} \rightarrow \delta_{ij}$.

Exercise 8: Consider the noninteracting Fermi system with ground state, $|\Psi_0\rangle$ in (34). Show that the retarded noninteracting many-body Green function is given in eigenstate basis as,

$$G_k(t - t') = -i\theta(t - t')e^{-iE_k(t-t')} \quad (46)$$

and in the general basis as,

$$G_{ij}(t - t') = -i\theta(t - t') \sum_k \phi_k(i)\phi_k^*(j)e^{-iE_k(t-t')} \quad (47)$$

with Fourier transform to frequency/energy domain,

$$G_{ij}(\omega) = \lim_{\eta \rightarrow 0^+} \sum_k \frac{\phi_k(i)\phi_k^*(j)}{\omega - E_k + i\eta}. \quad (48)$$

Note that the Green function in (48) contains the excitation energies of the system. The beauty of the Green function formalism for many-body systems is that it can be developed as a perturbation theory in the particle-interactions using a Dyson equation, and can yield the excitation spectrum also for interacting many-body systems. This excitation spectrum can then be compared to experimental data such as photo-electron or tunnel spectroscopy.